

BD-TRAFFIC GUARD

**ROBUST BANGLADESHI TRAFFIC SIGN DETECTION &
RECOGNITION UNDER REAL-WORLD ROAD CONDITIONS**

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Problem Gap: Why is Bangladeshi Traffic Sign Recognition Challenging?



Problem Gap: Why is Bangladeshi Traffic Sign Recognition Challenging?

- **Same traffic sign looks different under different weather and lighting.**
- **Different traffic signs may appear visually similar from a distance.**
- **Signs are often small, blurred, faded, or partially blocked.**
- **Existing models are rarely tested on real Bangladeshi road conditions.**
- **Robust real-time detection remains a challenging problem.**

MOTIVATION

Enhance Road Safety by accurately detecting and recognizing traffic signs in real-world conditions.

01

Support Autonomous Vehicles through reliable real-time traffic sign recognition.

02

Address Bangladesh-Specific Challenges such as blur, rain, fog, low light, occlusion, and faded signs.

03

Enable Practical Deployment on dashcams, mobile devices, and smart transportation systems.

04

Problem statement: Detecting and recognizing Bangladeshi traffic signs from real-world road images

Input: Road image containing one or more traffic signs

Output: Bounding box around each traffic sign & Traffic sign class label (e.g., Stop, No Parking, Speed Limit)

Constraints: Blur, Rain/fog, Low light, Occlusion, Small objects

Plan:

- Train on clean images
- Test on clean + degraded images

Evaluation Metrics:

Detection Metrics

- mAP@0.5
- mAP@0.5:0.95

Classification Metrics

- Precision
- Recall
- F1-score

Deployment Metrics

- FPS
- Inference Time
- Model Size

Robustness Metrics

- Robustness Drop

DATASET 1: BANGLADESH ROAD TRAFFIC SIGN DATASET

Name: [Bangladesh Road Traffic Sign Dataset \(2025\)](#), [Data in Brief \(ScienceDirect\)](#), [Zenodo](#)

Content: 10,259 real-world road images

Classes: 31 traffic sign classes

Format: Raw digital images (.jpg), Image annotation values (.txt)

Pixel scale: 640 × 640 pixels.

Key Features:

- Bounding-box annotations for object detection
- Includes variations in viewpoint, lighting, and background, Captured from different regions of Bangladesh

Purpose:

- Train and evaluate the traffic sign detection model
- Create clean train/validation/test splits
- Generate corrupted test sets for robustness evaluation

DATASET 2: BDROADSIGNS - IMAGE-BASED DATASET OF BD TRAFFIC SIGNS

Name: [BDRoadSigns: An Image-Based Dataset of Bangladeshi Traffic Signs for Supervised Classification Models](#)

Classes: 18 traffic sign classes

Format: Raw digital images (.jpg)

Pixel scale: 480 × 640 pixels .

Key Features:

- Cropped traffic sign images
- Suitable for image classification tasks
- Useful for transfer learning and additional validation

Purpose:

- Fine-tune traffic sign classification
- Compare classification performance
- Support two-stage detection + classification experiments

DATASET VISUALIZATION



LITERATURE REVIEW

Paper: Bangladeshi Traffic Sign Recognition and Classification using CNN with Different Kinds of Transfer Learning through a new (BTSRB) Dataset

Authurs: Md. Abu Sayeed, Md. Saiful Islam, Md Babul Islam, Piyush Kumar Pareek, Sabbir Ahmed, Tanbin Islam Rohan

Publication: 2023 (ICDCECE) | 979-8-3503-4745-6/23/\$31.00 ©2023 IEEE | DOI: 10.1109/ICDCECE57866.2023.10151254

Problem Statement:

Given an image of a cropped traffic sign, predict the class label ("Speed Limit 40", "No Overtaking")

Input: Cropped traffic sign image ; Output: Class label only

The model assumes that:

- All traffic sign classes are known during training (62 predefined classes)
- Input images are already cropped and centered (signs are pre-localized)
- Labeled examples are available for all classes (balanced dataset with 140 images per class)
- Test samples come from the same predefined label space

The model cannot detect signs in full road scenes, handle new/unseen traffic sign classes not in the training set, localize signs (find where they are in an image), handle images containing multiple signs and recognize signs under extreme degradations

LITERATURE REVIEW

**Paper: DSF-YOLO for Robust Multiscale Traffic Sign Detection Under Adverse Weather Conditions,
Scientific Reports, 2025.**

Authurs: Jun Li, QinWen Deng, WenXin Gao, Bing Yang, Lan Jial, Ju Zhou & HaiBo Pu

Publication: 08 July 2025, Scientific Reports Volume 15, Article number: 24550 (2025)

Problem Statement:

Given an image of a road scene, detect and classify traffic signs under adverse weather conditions (rain, fog, snow, shadow, glare, low brightness).

Input: Full road scene image : Output: Bounding Box + Class Label for each detected traffic sign

The model assumes that:

- **Traffic sign classes are known during training (45 classes from TT-100K)**
- **Weather conditions are simulated during training (not all real-world variations)**
- **Test samples come from the same dataset distribution**
- **Training data includes only specific traffic sign categories**

The model cannot detect traffic sign classes not seen during training, and recognize signs from regions with different sign designs (not trained on Bangladesh signs)

LITERATURE REVIEW

Paper: FUSED-Net: Detecting Traffic Signs with Limited Data

Authurs: Rahman, M.A., Asad, N.I., Omi, M.M.H., Hasan, M.B., Ahmed, S., & Kabir, M.H. (2024). IEEE Access

Publication: arXiv preprint (v2 published in January 2025)

Problem Statement:

Given an image of a road scene with limited labeled training data , detect and classify traffic signs with bounding box localization and class prediction, particularly for Bangladeshi traffic signs

Input: Full road scene image with limited labeled examples ; Output: Bounding Box + Class Label for each detected traffic sign

The model assumes that:

- **All traffic sign classes are known during training (15 predefined classes)**
- **Limited labeled examples are available (1-10 shots per class)**
- **Test samples come from the same predefined label space**

The model cannot detect completely novel traffic sign classes with zero training examples, handle new/unseen traffic sign classes not in the training set, detect signs under extreme weather conditions not simulated in training

FUTURE DIRECTIONS

Our Approach

Introduce a Bangladesh-specific robustness benchmark.

Evaluate performance under different degradation types like blur, rain, fog, glare, low light, and occlusion.

Improve small and distant traffic sign detection through multi-scale learning.

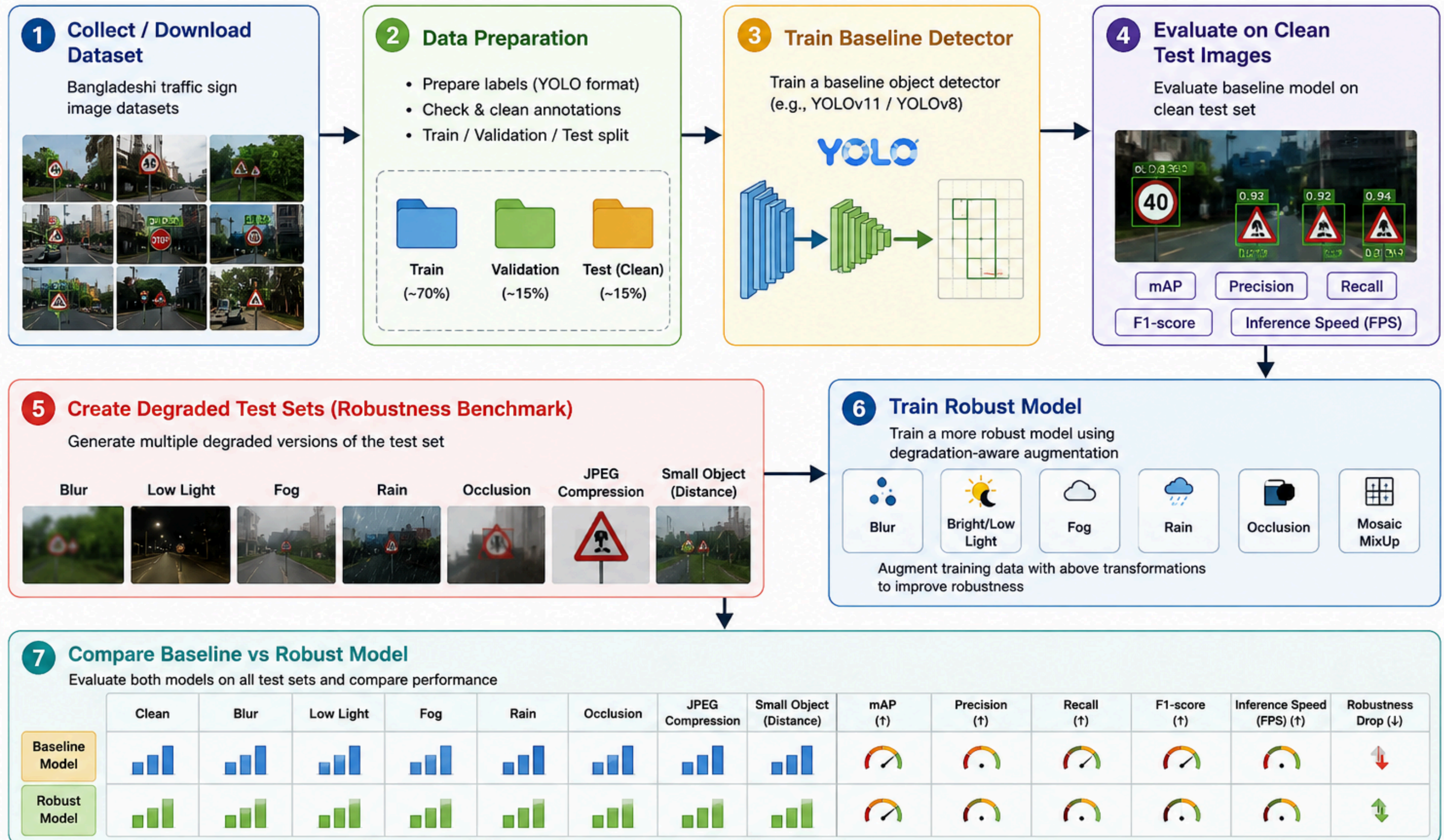
Compare standard training vs. degradation-aware augmentation.

Analyze class-wise robustness and failure cases by introducing Robustness Drop as key metric.

$\text{Robustness Drop} = \text{Clean mAP} - \text{Corrupted mAP}$

Optimize for real-time deployment on edge devices(dashcam).

BASIC PIPELINE



REFERENCES

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2. BanTracNet: Bangladeshi Trac Sign Recognition Using A Lightweight Deep Learning Approach
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5. Traffic Sign Detection and Recognition Using YOLO Object Detection Algorithm, Mathematics (MDPI), 2024.
6. BDRoadSigns: An Image-Based Dataset of Bangladeshi Traffic Signs for Supervised Classification Models
7. Bangladesh Road Traffic Sign Dataset (2025), Data in Brief (ScienceDirect), Zenodo
8. Ultralytics YOLO Documentation.
9. PyTorch Torchvision Faster R-CNN Documentation.

THANK YOU